FLORA T1 — crop and food protein function classifier (v1, descriptive-only)

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FLORA ("Functional Language Over Residue Annotations") is a protein **analysis** model from Kooya PH AI Research for proteins of Philippine-relevant crops and foods. This release is one task head: **T1, coarse protein function class** (enzyme, transport, storage, structural, other), trained as a LoRA adapter on top of a frozen open protein language model.

What is ours, and what is not. The base encoder is Meta's ESM-2 150M (MIT licence, Lin et al., 2023), downloaded and used unchanged. Ours are the benchmark (BIO-BENCH-001), the label scheme, the LoRA adapter and the classification head. This artifact contains the adapter and head only (184 tensors, 5.4 MB), not the base weights. We did not build a foundation model from scratch.

1 Intended use

- Classifying an **existing** protein sequence into one of five coarse function classes, for research on crop and food proteins.
- Inputs: one amino-acid sequence, up to 1,022 residues, standard ESM-2 alphabet.

2 Out of scope (by construction, not only by policy)

- Generating, designing or optimising protein sequences; inverse folding; any ligand, docking, binding-affinity or virtual-screening use. There is no sampling or generation path in this artifact or its loading code, and the training wrapper refuses to build or load a model that has one (governance ruling 565d09, analysis-only scope).
- Clinical, diagnostic, food-safety or regulatory decisions. T1 labels are research labels, not safety verdicts.
- Proteins outside plants, or sequences longer than 1,022 residues (these were dropped, never truncated, in the data).

3 Result (read the verdict line first)

Evaluated on the BIO-BENCH-001 T1 TEST split: 1,444 proteins in 498 sequence families, decontaminated against the training split. Three training seeds. Every interval is a 95% family-cluster bootstrap (B = 10,000) that resamples families, not single proteins.

Measure	Value
Headline: FLORA (LoRA) minus the frozen-encoder <cls> probe, mean paired delta in macro-F1	**+0.257**, 95% CI [+0.165, +0.433]
Per-seed delta	+0.241 / +0.271 / +0.260
FLORA macro-F1 per seed	0.729 / 0.747 / 0.742
Frozen-encoder <cls> probe macro-F1 per seed	0.488 / 0.476 / 0.482
Verdict	**DESCRIPTIVE-ONLY**

Why descriptive-only. The benchmark's pre-registration fixed a power floor before any result was read: if the minimum detectable difference exceeds the smallest effect of interest (0.03 macro-F1), T1 is reported descriptively and makes no confirmatory claim. Here the minimum detectable difference is 0.204, so the floor applies. The delta is large and its interval excludes zero, but under our own pre-registration it is a descriptive result, not a confirmed gain.

Against a fair baseline, FLORA shows no measured gain (BIO-BENCH v1.1, comparison C1; descriptive-only, the CI includes 0). The +0.257 above is measured against a deliberately simple probe on the <cls> token. Against the standard frozen baseline, a linear layer on the mean of the residue embeddings of the same frozen ESM-2, scored on the same TEST set with the same protocol, the paired delta is **-0.049**, 95% CI [−0.136, +0.105] (pre-declared descriptive-only). Seed-mean 5-class macro-F1: FLORA 0.739, mean-pooled probe 0.788. The whole difference sits in the 6-family storage class (storage F1 0.274 for FLORA vs 0.625 for the mean-pooled probe, which does not over-call storage); without storage, FLORA is 0.856 vs 0.829 (point estimates only).

Per-class F1 for the released checkpoint (median seed, seed 2):

Class	F1	TEST proteins
structural	0.931	75
enzyme	0.905	562
other	0.836	578
transport	0.810	218
storage	0.230	11 (6 families)

Macro-F1 over the four classes with adequate support (storage excluded): 0.870.

4 Limitations (please read before citing a number)

1. **The headline comparator is a weak baseline; against the fair one there is no measured gain.** The +0.257 is over one linear layer on the <cls> token. Against the standard mean-pooled frozen probe (pre-registered v1.1 C1, run on the same TEST set), FLORA's 5-class delta is −0.049, 95% CI [−0.136, +0.105]: no measured gain (descriptive-only; the interval includes zero). We make no state-of-the-art claim.
2. **Storage is not measurable here.** The 151 storage proteins fall into only 30 families, and only 6 are in TEST (sizes 3, 2, 2, 2, 1, 1; effective ~5.3 clusters by Kish). Storage F1 is reported but should not be relied on. This class widens the minimum detectable difference, but the verdict does not depend on it: in the registered power analysis on the same TEST set, the 4-class macro without storage still has a

minimum detectable difference of 0.153, about five times the 0.03 threshold. The released checkpoint also over-predicts storage (50 predicted against 11 true in TEST); 40 of the 43 wrong storage calls are one germin-like (cupin-fold) family.

3. **Labels are single-label by precedence.** A protein with several function signals receives one label by a fixed, pre-registered order (storage > transport > structural > enzyme > other). A T1 label is the precedence-resolved class, not a claim that the protein has only that function.
4. **Pretraining overlap is not yet reported.** ESM-2 was pretrained on UniRef50 release 2021_04, which contains these reviewed proteins or their homologs. Both arms share that exposure; it is expected to inflate the absolute macro-F1 values more than the delta, but this has NOT been measured, so the delta is not guaranteed unbiased. The pretraining-overlap figure is pending.
5. **Scope.** Reviewed UniProtKB/Swiss-Prot proteins from 14 crop and food plants, with rice the largest share. Not validated on other organisms or on unreviewed sequences.
6. **Scores are not calibrated.** On TEST, accuracy is 0.851 while the mean top score is 0.915 (ECE 0.068, 10 bins); storage calls scoring above 0.9 were right 3 of 15 times (20%). Do not read a score as a probability of being correct. Read instead how often the predicted class was right on TEST: structural 67/69, enzyme 514/574, other 471/549, transport 170/202, storage 7/50 (pinned table `BIO-RELEASE-001-FLORA-T1-class-precision.json`).

5 Training data and procedure

Data provenance. Benchmark data: 7,743 reviewed UniProtKB/Swiss-Prot proteins (CC BY 4.0; The UniProt Consortium) from 14 crop species, drawn on 2026-10-23 through UniProt's public REST API (robots.txt-compliant). Toxin (KW-0800), virulence (KW-0843) and viral entries were excluded in the query and re-screened on every row. T1 labels come from UniProt keyword / GO (is_a subtree) / EC annotations, assigned by a fixed precedence. The data was split by 30%-identity MMseqs2 family clusters (MMseqs2 18-8cc5c, MIT), and after a search-based homology removal no TEST protein has a >=30% homolog in TRAIN (1,444 TEST proteins in 498 families). No sequences or training data ship with this adapter.

- **Source:** UniProtKB/Swiss-Prot reviewed entries (CC BY 4.0) for 14 crop and food plant taxa: peanut, wheat, maize, rice, potato, soybean, coffee, cassava, banana, cacao, tomato, mung bean, pineapple, coconut.
- **Safety screen:** entries carrying UniProt keyword KW-0800 (Toxin) or KW-0843 (Virulence), viral entries, and non-plant organisms filed under a crop taxon were excluded before any labelling.
- **Labels:** each class is defined by UniProt query-set membership on reviewed annotations (storage KW-0758; transport KW-0813 or GO:0005215; structural GO:0005198; enzyme EC number or GO:0003824; other = remaining), then resolved by the precedence above.
- **Split and leakage control:** MMseqs2 clustering at 30% identity; TRAIN and TEST are disjoint family sets; 101 TEST proteins that still had a 30%-identity homolog in TRAIN were removed before scoring, and a re-check found none left.
- **Sizes:** 7,642 proteins after the length filter and decontamination; TRAIN 6,198 / TEST 1,444.
- **Training:** LoRA rank 8, alpha 16, dropout 0.1 on attention query/key/value, plus a classification head; base frozen; batch 16, 3 epochs, fp16 autocast, gradient checkpointing; about 25 minutes per seed on one NVIDIA T4.

6 How to load

```python
from transformers import AutoTokenizer, EsmForSequenceClassification
from peft import PeftModel

base = "facebook/esm2_t30_150M_UR50D"
rev = "a695f6045e2e32885fa60af20c13cb35398ce30c"
tok = AutoTokenizer.from_pretrained(base, revision=rev)
model = EsmForSequenceClassification.from_pretrained(base, revision=rev, num_labels=5)
model = PeftModel.from_pretrained(model, "<this-repo>")
labels = ["enzyme", "transport", "storage", "structural", "other"]
```

7 Citation and authorship

Stan Lee V. Natividad · Kooya PH AI Research. Base model: Lin, Z. et al. (2023), "Evolutionary-scale prediction of atomic-level protein structure with a language model", *Science* 379, 1123–1130. Data: The UniProt Consortium, UniProtKB/Swiss-Prot, CC BY 4.0.